

Canonical-inner-product Toeplitz operators on Fock–Sobolev type spaces

An identity-plus-compact transfer theorem

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Status. Candidate full result, likely valid. Theorems A–C of Cho–Isralowitz–Joo transfer to the canonical inner product, including the canonical Berezin-transform characterization for every Schatten exponent $0 < t < \infty$. The proof is a direct operator-theoretic transfer, augmented by an exact canonical-kernel localization lemma.

1 Source question

The source is H. R. Cho, J. Isralowitz and J.-C. Joo, *Toeplitz operators on Fock–Sobolev type spaces*, arXiv:1307.4829, later published in *Integral Equations and Operator Theory* 82 (2015), 1–32 [1]. Section 5 asks whether its Theorems A, B and C remain true when the Toeplitz operator is defined from the reproducing kernel for the natural inner product, and specifically asks for a direct operator-theoretic transfer between the two equivalent Hilbert inner products. The decisive source passage is reproduced below.

5. REMARKS AND OPEN PROBLEMS

As was remarked in the introduction, it would be interesting to know if one can prove Theorems A, B, and C for Toeplitz operators with nonnegative Borel measure symbols that are defined in terms of the reproducing kernel of F_{α}^2 with respect to the inner product (1.1). Note that as usual, the main difficulty in doing this is obtaining the necessary estimates for the reproducing kernel defined with respect to the inner product (1.1). Furthermore, it would be quite interesting to know if there is a direct operator theoretic argument that allows one to directly transfer information about Toeplitz operators defined with respect to one Hilbert space inner product to a Toeplitz operator defined with respect to a different Hilbert space inner product (where the norms these inner products generate are equivalent.) In particular, it would be interesting to know if one can employ operator theoretic techniques to prove Theorems A, B, or C (or even portions of these theorems) for Toeplitz operators defined with respect to $\langle \cdot, \cdot \rangle_{L_{\alpha}^2}$ directly from Theorems A, B, or C, respectively.

Figure 1: The open problem in Section 5, page 31 of arXiv:1307.4829v2.

2 Definitions and statement

Write

$$dv_\alpha(z) = (1 + |z|)^{-\alpha} dv(z), \quad \|f\|_{F_\alpha^p}^p = \int_{\mathbb{C}^n} |f(z)|^p e^{-p|z|^2/2} dv_\alpha(z).$$

The canonical Hilbert inner product on F_α^2 is

$$C(f, g) := \int_{\mathbb{C}^n} f \bar{g} e^{-|z|^2} (1 + |z|)^{-\alpha} dv.$$

Let K_z^c be its reproducing kernel. For a positive locally finite Borel measure μ , the canonical Toeplitz operator is initially defined on polynomials by

$$\mathcal{T}_\mu f(z) = \int_{\mathbb{C}^n} f(w) \overline{K_z^c(w)} e^{-|w|^2} (1 + |w|)^{-\alpha} d\mu(w). \quad (2.1)$$

Equivalently,

$$C(\mathcal{T}_\mu f, g) = C_\mu(f, g) := \int_{\mathbb{C}^n} f \bar{g} e^{-|z|^2} (1 + |z|)^{-\alpha} d\mu. \quad (2.2)$$

Let M be the modified inner product used in [1]. If $\alpha \leq 0$,

$$M(f, g) = \int f \bar{g} e^{-|z|^2} |z|^{-\alpha} dv.$$

If $\alpha > 0$, let P be the Taylor projection onto homogeneous degrees at most $\alpha/2$, let $Q = I - P$, and put

$$M(f, g) = \int P f \overline{P g} e^{-|z|^2} dv + \int Q f \overline{Q g} e^{-|z|^2} |z|^{-\alpha} dv. \quad (2.3)$$

The source operator $\mathcal{T}_\mu$ represents the analogous form M_μ .

Fix a sufficiently small radius $r > 0$ and an r -lattice $\{a_j\} \subset \mathbb{C}^n$. The choice of lattice and radius is immaterial for the conditions below.

Theorem 1 (Canonical operator/geometric criteria). *Let $\alpha \in \mathbb{R}$, and let $\mu \geq 0$ satisfy the natural dense-domain condition for (2.1). Then:*

1. *For every $1 \leq p < \infty$, $\mathcal{T}_\mu$ extends boundedly on F_α^p if and only if*

$$\sup_{z \in \mathbb{C}^n} \mu(B(z, r)) < \infty. \quad (2.4)$$

2. *For every $1 \leq p < \infty$, $\mathcal{T}_\mu$ is compact on F_α^p if and only if*

$$\mu(B(z, r)) \longrightarrow 0 \quad (|z| \rightarrow \infty). \quad (2.5)$$

3. *For every $0 < t < \infty$, as an operator on the canonical Hilbert space F_α^2 ,*

$$\mathcal{T}_\mu \in \mathcal{S}_t \iff \mu(B(\cdot, r)) \in L^t(dv) \iff \{\mu(B(a_j, r))\}_j \in \ell^t. \quad (2.6)$$

Moreover, if $\tilde{\mu}_c(z) = C(\mathcal{T}_\mu k_z^c, k_z^c)$ is the canonical Berezin transform, then it is bounded exactly under (2.4), and it vanishes at infinity exactly under (2.5). For every $0 < t < \infty$,

$$\tilde{\mu}_c \in L^t(dv) \iff \mathcal{T}_\mu \in \mathcal{S}_t. \quad (2.7)$$

Thus all clauses of source Theorems A, B and C survive the canonical inner product.

3 Proof intuition

Equivalent Hilbert norms alone do not preserve Toeplitz operators, because a Toeplitz operator also remembers the symbol form. Here the two forms have much more structure. Outside a fixed ball their weights differ by

$$q(z) = \left(\frac{|z|}{1 + |z|} \right)^\alpha, \quad q(z) \longrightarrow 1,$$

while for $\alpha > 0$ all terms involving the low Taylor projection P have finite rank. Contributions from a fixed ball are better still: analytic Taylor approximation makes them belong to every Schatten class.

The operator S changing the modified inner product into the canonical one therefore satisfies $S = I + K$ with K compact on every F_α^p . Radiality makes S diagonal with strictly positive diagonal entries, so it is injective; Fredholm theory makes it invertible. After multiplication by S , the canonical Toeplitz operator is the source Toeplitz operator with symbol $q\mu$, modulo the local and finite-rank corrections. Since $q \asymp 1$ off a fixed ball, all ball-mass criteria transfer unchanged. To handle the quasi-Banach Berezin clause, the canonical weight is smoothed only on a fixed ball. Generalized-Fock kernel localization for the smoothed weight transfers back through a compactly supported norm change and yields exponential off-diagonal decay for the *exact* canonical kernel.

4 The transfer argument

Lemma 2 (Compactly supported forms are super-smoothing). *Suppose σ is a finite positive measure supported in a fixed ball. The operator representing C_σ , with respect to either of the two equivalent Hilbert structures, belongs to $\mathcal{S}_t$ for every $t > 0$. The corresponding operator on F_α^p is compact for every $1 \leq p < \infty$. The same conclusion holds for the high-high modified form containing $|z|^{-\alpha}$, after applying Q when $\alpha > 0$.*

Proof. Let Π_N be the Taylor projection onto total degree at most N . Its rank is $O(N^n)$. On a fixed ball, the normalized monomial estimates for a Gaussian Fock norm give

$$\sup_{|z| \leq R} |(I - \Pi_N)f(z)| \leq A_R \frac{B_R^N}{\sqrt{N!}} \|f\|_{F_\alpha^2}. \quad (4.1)$$

For the high part, multiplication by $|z|^{-\alpha/2}$ is harmless once $N > \alpha/2$, because every omitted term vanishes to higher order at zero. Consequently the approximation numbers of the local embedding decay faster than any power after a rank $O(N^n)$ truncation. The embedding, and hence its positive square, lies in every Schatten class. The same Taylor truncations converge in operator norm on F_α^p , proving compactness. $\square$

Define S first on polynomials by

$$M(Sf, g) = C(f, g). \quad (4.2)$$

Lemma 3 (The norm-change operator). *For every $1 \leq p < \infty$, S extends to a bounded invertible operator on F_α^p , and*

$$S - I \text{ is compact on } F_\alpha^p. \quad (4.3)$$

On F_α^2 , the compact difference belongs to every Schatten class after removal of the exterior source- Toeplitz term described below. The kernels satisfy

$$K_z^c = S^{-1} K_z^m. \quad (4.4)$$

Proof. Take a cutoff at $|z| = R \geq 1$. On the exterior high-degree block, $C - M$ is the modified Toeplitz form with scalar symbol $q - 1$, where $q = (|z|/(1 + |z|))^\alpha$. This symbol is bounded there and tends to zero. The compactness part of Theorem B in [1], applied to $|q - 1|dv$, shows that this exterior difference is compact on every F_α^p . The interior difference is compact by Lemma 2. When $\alpha > 0$, the remaining blocks contain P and hence have finite rank. This proves (4.3) and boundedness.

Both radial forms make the monomials orthogonal. Thus S is diagonal on homogeneous polynomials, and every diagonal coefficient is a ratio of two strictly positive moments. Hence S is injective on every F_α^p . Since $S = I + \text{compact}$, it is Fredholm of index zero and therefore invertible. Finally, for every f ,

$$f(z) = M(f, K_z^m) = M(Sf, K_z^c) = M(f, SK_z^c),$$

which gives (4.4). □

Lemma 4 (Toeplitz transfer modulo negligible blocks). *Assume $\sup_z \mu(B(z, r)) < \infty$. Put*

$$d\nu(z) = \mathbf{1}_{\{|z| > R\}} \left(\frac{|z|}{1 + |z|} \right)^\alpha d\mu(z).$$

Then, on every F_α^p ,

$$S\mathcal{T}_\mu = T_\nu + E_\mu, \tag{4.5}$$

where E_μ is compact. On F_α^2 , $E_\mu \in \bigcap_{t>0} \mathcal{S}_t$. In particular, for every $t > 0$,

$$\mathcal{T}_\mu \in \mathcal{S}_t \iff T_\nu \in \mathcal{S}_t. \tag{4.6}$$

Proof. For $\alpha \leq 0$, the exterior canonical form is exactly the modified form with symbol ν , because

$$|z|^{-\alpha} \left(\frac{|z|}{1 + |z|} \right)^\alpha = (1 + |z|)^{-\alpha}.$$

For $\alpha > 0$, the same identity gives equality of the high-high blocks. Every discrepancy involving P has finite rank. The part supported in $B(0, R)$ is covered by Lemma 2. Equation (4.5) follows from the defining identity $M(S\mathcal{T}_\mu f, g) = C_\mu(f, g)$. The Schatten assertion follows because the error is in every $\mathcal{S}_t$ and left multiplication by the bounded invertible operator S preserves every Schatten ideal, including the quasi-ideals with $t < 1$. □

Proof of Theorem 1. We first prove necessity of (2.4) without any kernel estimate. Put $E_a(z) = e^{z \cdot \bar{a}}$. Gaussian localization gives

$$\|E_a\|_{F_\alpha^p} \asymp e^{|a|^2/2} (1 + |a|)^{-\alpha/p}. \tag{4.7}$$

Use the analogous conjugate-exponent normalization, with the unweighted Fock F^∞ norm when $p = 1$. If f_a, g_a are the two normalized exponentials, then (2.2) and $|w - a| < r$ give

$$|C(\mathcal{T}_\mu f_a, g_a)| = C_\mu(f_a, g_a) \gtrsim \mu(B(a, r)). \tag{4.8}$$

The canonical pairing is bounded by Holder's inequality, so boundedness of $\mathcal{T}_\mu$ makes the right side uniformly bounded.

Conversely, (2.4) implies that ν in Lemma 4 has uniformly bounded ball masses. Theorem A of [1] makes T_ν bounded on every F_α^p . Equation (4.5) and invertibility of S prove boundedness of $\mathcal{T}_\mu$.

If (2.5) holds, the ball masses of ν vanish, so source Theorem B and (4.5) make $\mathcal{T}_\mu$ compact. Conversely, compactness first gives boundedness and hence (2.4). Then (4.5) makes T_ν compact;

source Theorem B implies that the ball masses of ν , and therefore those of μ outside a fixed ball, vanish. This proves (2.5).

Finally, source Theorem C says

$$T_\nu \in \mathcal{S}_t \iff \{\nu(B(a_j, r))\} \in \ell^t.$$

Since $q \asymp 1$ on the exterior and only finitely many lattice cells meet the omitted ball, this sequence lies in ℓ^t exactly when $\{\mu(B(a_j, r))\}$ does. Lemma 4 proves (2.6). The equivalence with $\mu(B(\cdot, r)) \in L^t$ is the standard bounded-overlap lattice discretization (valid also for $0 < t < 1$). $\square$

5 Canonical Berezin transform

Put

$$\phi_c(z) = \frac{|z|^2}{2} + \frac{\alpha}{2} \log(1 + |z|), \quad e^{-2\phi_c(z)} = e^{-|z|^2} (1 + |z|)^{-\alpha}.$$

Lemma 5 (Exact canonical off-diagonal decay). *There are constants $C, \varepsilon > 0$, depending only on n and α , such that*

$$|K_z^c(w)| e^{-\phi_c(z) - \phi_c(w)} \leq C e^{-\varepsilon|z-w|}, \quad (5.1)$$

and

$$K_z^c(z) e^{-2\phi_c(z)} \asymp 1. \quad (5.2)$$

Consequently,

$$|k_z^c(w)|^2 e^{-2\phi_c(w)} \leq C e^{-2\varepsilon|z-w|}. \quad (5.3)$$

Proof. We first remove the harmless nonsmoothness of ϕ_c at the origin without changing it at infinity. For $s > 0$, write $\phi_c(z) = F_0(|z|^2)$. If $r = \sqrt{s}$, the tangential and radial eigenvalues of the complex Hessian are, up to the fixed normalization of dd^c ,

$$F_0'(s) = \frac{1}{2} + \frac{\alpha}{4r(1+r)}, \quad F_0'(s) + sF_0''(s) = \frac{1}{2} + \frac{\alpha}{8r(1+r)^2}. \quad (5.4)$$

They are bounded above and below by positive constants for r sufficiently large. A standard radial interpolation now gives $\psi \in C^2(\mathbb{C}^n)$, equal to ϕ_c outside a ball, such that

$$c\omega_0 \leq dd^c\psi \leq C\omega_0. \quad (5.5)$$

For completeness, set $g_0(s) = sF_0'(s)$ on a sufficiently large exterior interval. Extend it smoothly to $s = 0$, with $g(s) = \lambda s$ near zero, so that $0 < c \leq g'(s) \leq C$, while retaining $g = g_0$ on the exterior. This is possible because both $g_0'(s)$ and $g_0(s)/s$ have such bounds there. Integrating $\Psi'(s) = g(s)/s$ and choosing the additive constant at the joining radius gives $\psi(z) = \Psi(|z|^2)$; its Hessian eigenvalues are $g(s)/s$ and $g'(s)$, proving (5.5).

Let H_ψ be the generalized Fock Hilbert space with weight $e^{-2\psi}$, and let L_z be its reproducing kernel. The generalized Fock kernel estimate [3, Lemmas 2.1–2.2] gives

$$|L_z(w)| e^{-\psi(z) - \psi(w)} \leq C e^{-\varepsilon|z-w|}, \quad L_z(z) e^{-2\psi(z)} \asymp 1. \quad (5.6)$$

Since $\psi = \phi_c$ off a compact set, the two Hilbert norms differ by a bounded compactly supported multiplier. More precisely, with

$$h = e^{2(\psi - \phi_c)} - 1,$$

we have

$$C(f, g) = \langle Af, g \rangle_\psi, \quad A = I + T_h^\psi, \quad (5.7)$$

where $T_h^\psi = P_\psi M_h$. The operator A is positive and boundedly invertible, while T_h^ψ is a compactly supported operator of the type in Lemma 2. Reproducing gives

$$K_z^c = A^{-1}L_z = L_z + BL_z, \quad B = A^{-1} - I = -A^{-1}T_h^\psi. \quad (5.8)$$

Projection contractivity, compact support of h , and (5.6) imply

$$\|T_h^\psi L_z\|_\psi \leq \|hL_z\|_{L^2(e^{-2\psi})} \leq Ce^{\psi(z)}e^{-\varepsilon|z|}. \quad (5.9)$$

Point evaluation and the diagonal estimate in (5.6) therefore give

$$|(BL_z)(w)| \leq Ce^{\psi(z)+\psi(w)}e^{-\varepsilon|z|}. \quad (5.10)$$

The operator B is self-adjoint, so its kernel is Hermitian; interchanging z, w gives the same bound with $e^{-\varepsilon|w|}$. Taking the better bound and using $\max\{|z|, |w|\} \geq |z - w|/2$ shows that the correction in (5.8) has the exponential off-diagonal bound in (5.1). Combining this with (5.6), and using that $\psi - \phi_c$ is bounded, proves (5.1).

Equivalent Hilbert norms give comparable squared norms for every point evaluation. Thus (5.6) implies (5.2), and division by $K_z^c(z)$ gives (5.3). $\square$

We record why no explicit formula for K^c is needed for the bounded and vanishing clauses. The normalized modified kernels k_z^m converge weakly to zero as $|z| \rightarrow \infty$. Lemma 3 therefore yields

$$\|(S^{-1} - I)k_z^m\|_M \rightarrow 0. \quad (5.11)$$

The source near-diagonal lower estimate for K^m , together with point evaluation and (4.4), implies that for some $r_0, c > 0$,

$$|K_z^c(w)| \geq c\sqrt{K_z^c(z)K_w^c(w)} \quad \text{when } |w - z| < r_0 \text{ and } |z| \text{ is large.} \quad (5.12)$$

Indeed, the difference in (5.11), evaluated at w , is $o(\sqrt{K_w^m(w)})$, uniformly in w , while the source kernel has a fixed lower constant on the ball. Equivalent Hilbert norms give

$$K_z^c(z) \asymp K_z^m(z) \asymp (1 + |z|)^\alpha e^{|z|^2}. \quad (5.13)$$

It follows from (5.12)–(5.13) that

$$\tilde{\mu}_c(z) \gtrsim \mu(B(z, r_0)) \quad (5.14)$$

for large $|z|$. Thus boundedness or vanishing of the canonical Berezin transform forces the corresponding ball condition. The converses follow from boundedness or compactness of $\mathcal{T}_\mu$; normalized canonical kernels are weakly null at infinity.

For $t \geq 1$, positivity and Jensen's inequality give

$$\tilde{\mu}_c(z)^t \leq C((\mathcal{T}_\mu)^t k_z^c, k_z^c).$$

The canonical continuous-frame measure

$$K_z^c(z)e^{-|z|^2}(1 + |z|)^{-\alpha} dv(z)$$

is comparable with $dv(z)$ by (5.3). Integration proves $\|\tilde{\mu}_c\|_{L^t}^t \lesssim \|\mathcal{T}_\mu\|_{\mathcal{S}_t}^t$. The reverse implication follows from (5.14), lattice discretization, and (2.6).

It remains only to prove the forward implication when $0 < t < 1$. Suppose $\mathcal{T}_\mu \in \mathcal{S}_t$. By (2.6), if $m_j = \mu(B(a_j, r))$, then $\{m_j\} \in \ell^t$. Choose a measurable partition $\{D_j\}$ of $\mathbb{C}^n$ with $D_j \subset B(a_j, r)$. Lemma 5 gives

$$\tilde{\mu}_c(z) \leq C \int_{\mathbb{C}^n} e^{-2\varepsilon|z-w|} d\mu(w) \leq C_r \sum_j m_j e^{-\varepsilon|z-a_j|}. \quad (5.15)$$

Since $0 < t < 1$, subadditivity and translation invariance yield

$$\int_{\mathbb{C}^n} \tilde{\mu}_c(z)^t dv(z) \leq C_{r,t} \sum_j m_j^t \int_{\mathbb{C}^n} e^{-t\varepsilon|z-a_j|} dv(z) \leq C'_{r,t} \sum_j m_j^t < \infty. \quad (5.16)$$

This proves (2.7) throughout the quasi-Banach range and completes all claims.

6 Verification, scope, and novelty

- The proof uses only the source Theorems A–C, its modified-kernel near-diagonal estimate, elementary Fredholm theory, Taylor approximation, and the generalized-Fock kernel estimate of Isralowitz–Virtanen–Wolf.
- The potentially singular factor at the origin is always placed after the high Taylor projection Q ; Lemma 2 explicitly uses the resulting order of vanishing.
- All assertions involving arbitrary p concern boundedness and compactness. Schatten ideals concern the canonical Hilbert space F_α^2 .
- The exact canonical kernel is proved exponentially localized by a compactly supported smoothing of the weight. The correction kernel decays in one variable by support localization and in the other by self-adjointness.
- A bounded search on 27 August 2026 used the exact source id/title and the canonical/natural-inner-product Fock–Sobolev Toeplitz keywords. No matching transfer result was located. A 2023 paper still uses the modified inner product and repeats the canonical-kernel difficulty; this is evidence of continued relevance, not proof of novelty.

Human-review recommendation. Focused expert review should audit the local-form compactness in Lemma 2, the $\alpha > 0$ block identity (4.5), and the radial interpolation/self-adjoint two-variable decay in Lemma 5.

References

- [1] H. R. Cho, J. Isralowitz and J.-C. Joo, *Toeplitz operators on Fock–Sobolev type spaces*, Integral Equations Operator Theory 82 (2015), 1–32; [arXiv:1307.4829](#).
- [2] H. R. Cho, B. R. Choe and H. Koo, *Fock–Sobolev spaces of fractional order*, Potential Analysis 43 (2015), 199–240; [arXiv:1307.2670](#).
- [3] J. Isralowitz, J. A. Virtanen and L. Wolf, *Schatten class Toeplitz operators on generalized Fock spaces*, J. Math. Anal. Appl. 421 (2015), 329–337; [arXiv:1402.2567](#).